FLIGHT ANALYTICS · DATA SCIENCE PROJECT

Flight Price Analysis & Prediction

What factors determine flight ticket prices — and can we predict them?

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INTRODUCTION

Project Objective

Analyze flight data to understand the factors that affect ticket prices, and build a machine learning model capable of predicting them.

A structured, three-stage data science workflow — from raw data to predictive insight.

Clean & Prepare

01

Inspect, de-duplicate, and validate the raw flight dataset.

Explore Visually

02

Use charts to uncover patterns between price and flight features.

Model & Predict

03

Train a Linear Regression model to estimate flight ticket prices.

DATA LOADING & INSPECTION

Dataset Overview

BEFORE CLEANING

10,683

rows × 14 columns

AFTER REMOVING DUPLICATES

10,461

rows × 14 columns

AFTER ENCODING

10,461

rows × 26 columns

Inspection Steps

- Inspected the raw dataset structure
- Checked dimensions (rows & columns)
- Verified data types across all fields
- Generated descriptive statistics

Column Composition

3 categorical fields (object)

11 numeric fields (int64)

0 missing values found

DATA QUALITY

Data Cleaning Process

Missing Values

Checked the full dataset for missing values.

RESULT

None found.

Duplicate Records

Checked for and removed duplicate flight entries.

RESULT

10,683 → 10,461 rows.

Categorical Variables

Inspected Airline, Source, Destination and Total_Stops.

RESULT

Ready for encoding.

Price Outliers

Detected potential outliers using the IQR method.

RESULT

Retained — may be valid high fares.

PREPARING THE DATA

Feature Engineering

Raw fields were transformed into more meaningful, model-ready features.

01

Journey_Date

Combined Date, Month, and Year columns into a single unified journey date field.

02

Day_of_Week

Extracted the day of the week from the journey date to capture weekly patterns.

03

Duration_Total_Minutes

Converted flight duration (hours + minutes) into one continuous minute value.

04

Arrival_Time_Minutes

Converted arrival hours and minutes into total minutes for consistency.

END-TO-END PIPELINE

Data Preprocessing Workflow

1 Missing Values

Problem Could bias the model
Solution Checked with isnull().sum()
Result None found

2 Duplicate Records

Problem Redundant flight entries
Solution duplicated() + drop_duplicates()
Result 10,683 → 10,461 rows

3 Categorical Variables

Problem Text values unusable by regression
Solution Ordinal + One-Hot encoding
Result 26 numeric features

4 Feature Engineering

Problem Info split across columns
Solution Combined date & time fields
Result Cleaner, continuous inputs

5 Price Outliers

Problem Extreme values may skew fit
Solution Identified via IQR method
Result Retained as valid fares

6 Redundant Columns

Problem Original fields no longer needed
Solution Dropped after engineering
Result Only relevant features kept

Result: a clean, ML-ready dataset — no missing values, duplicates removed, categorical features encoded, and only relevant engineered features retained.

Price Distribution & Airline Comparison

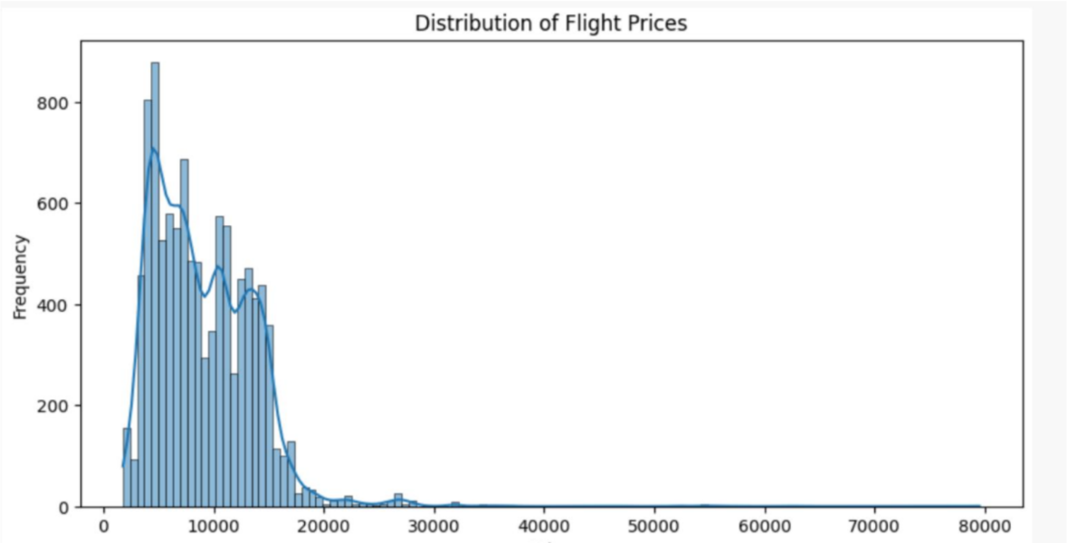

Figure 1 · Price Distribution

Most tickets cluster between ₹5,000–15,000; the distribution is right-skewed, with a long tail of higher-priced fares.

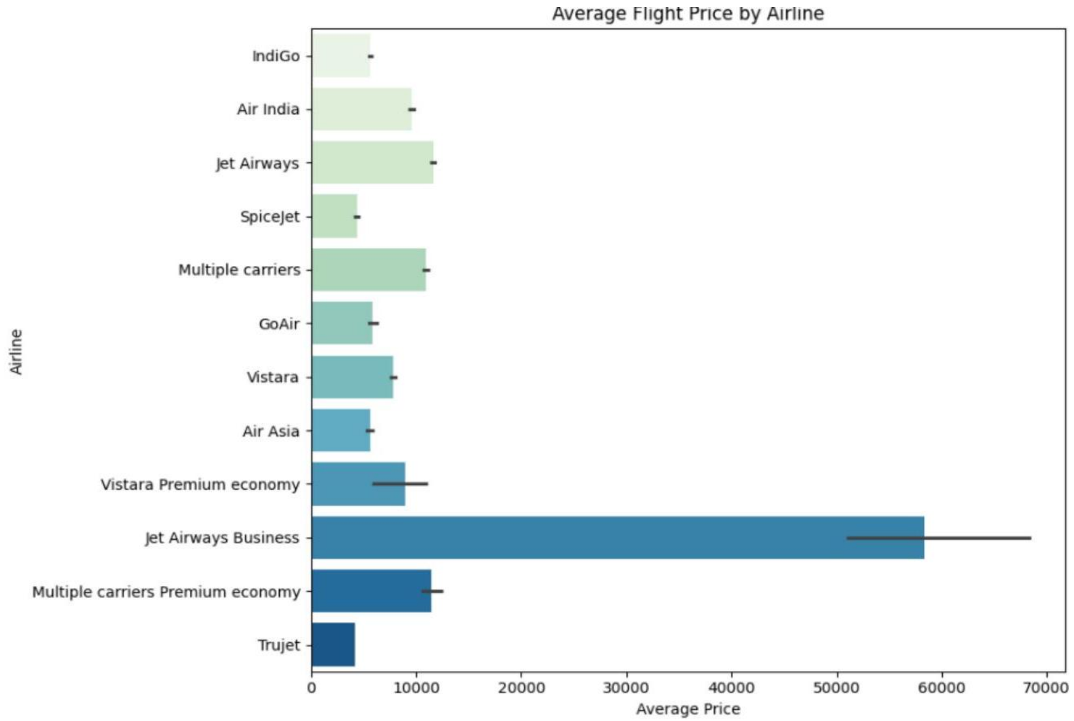

Figure 2 · Average Price by Airline

Airline choice is strongly associated with price — premium and business-class carriers command the highest average fares.

Stops & Flight Duration Impact

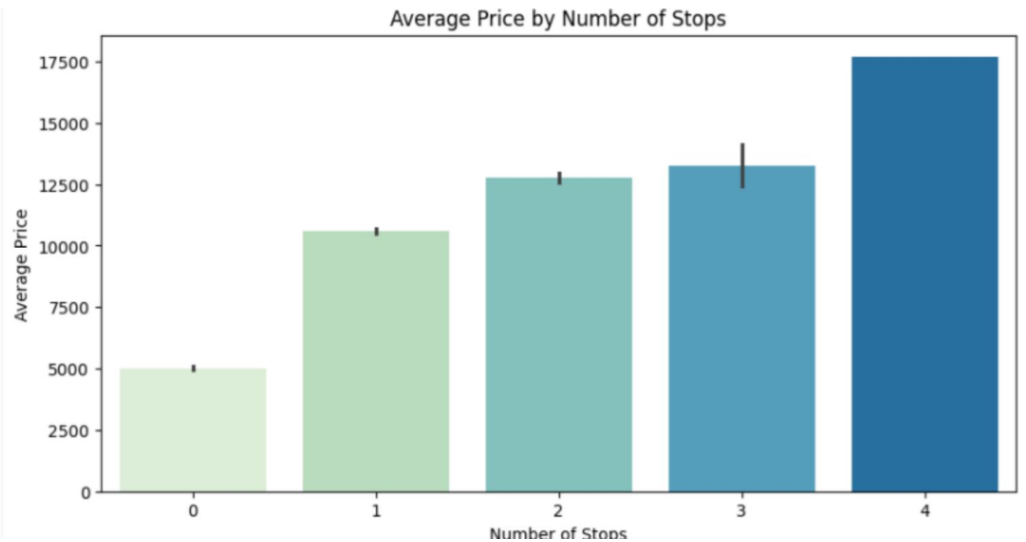

Figure 3 · Average Price by Number of Stops

Prices rise steadily with each additional stop — non-stop flights average ~₹5,000, versus ~₹17,700 for 4-stop itineraries.

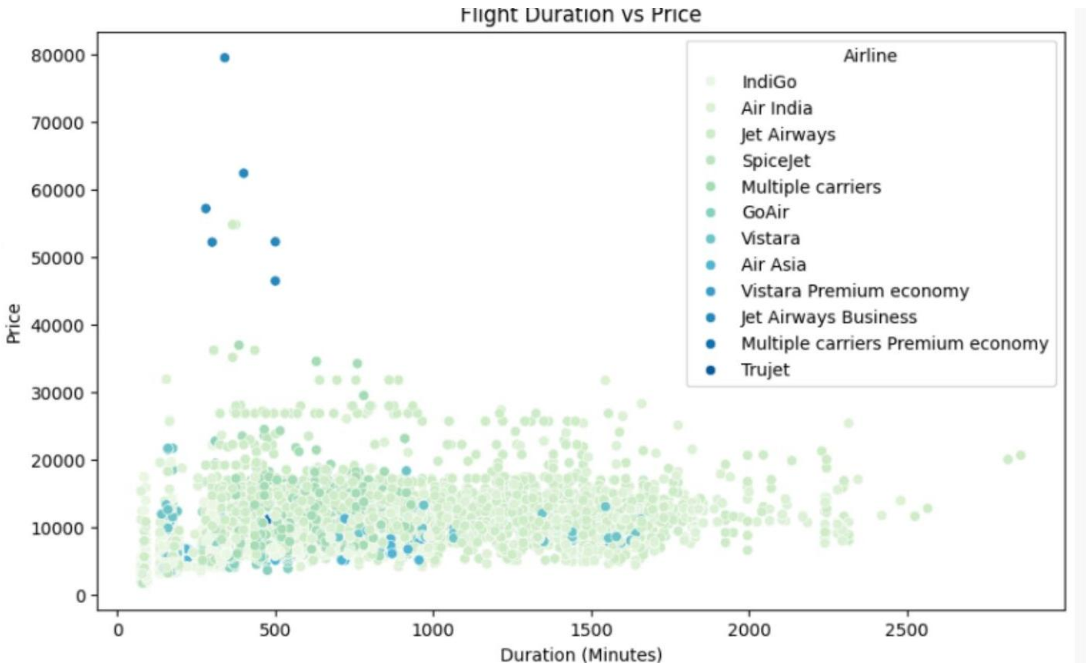

Figure 4 · Flight Duration vs Price

Duration shows a relationship with ticket price, making it a strong candidate feature for the prediction model.

Departure & Destination Comparison

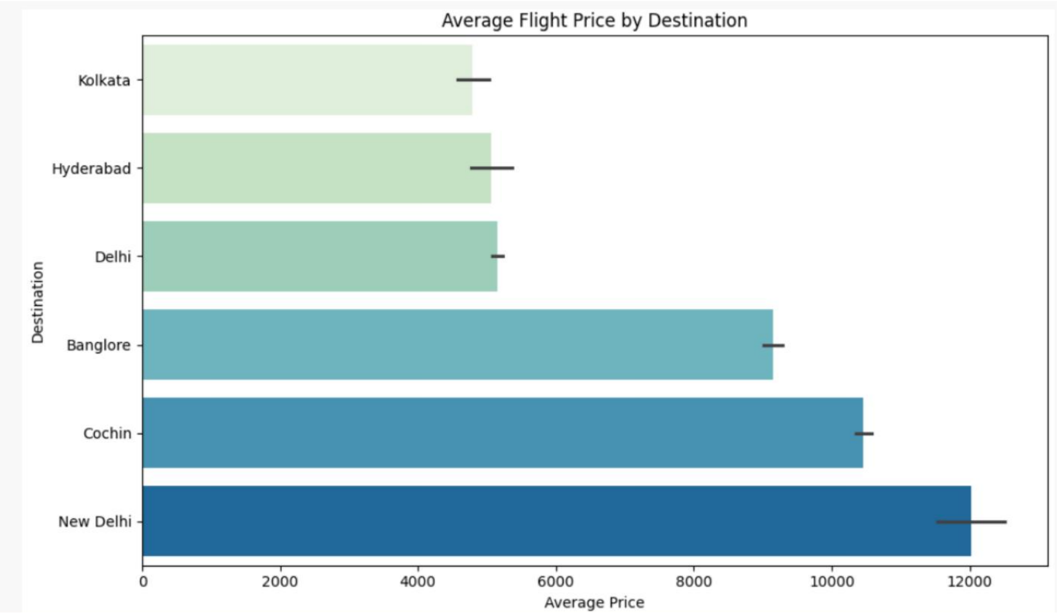

Figure 5 · Average Price by Destination

Average ticket prices differ noticeably across destinations, with New Delhi and Cochin routes commanding the highest fares.

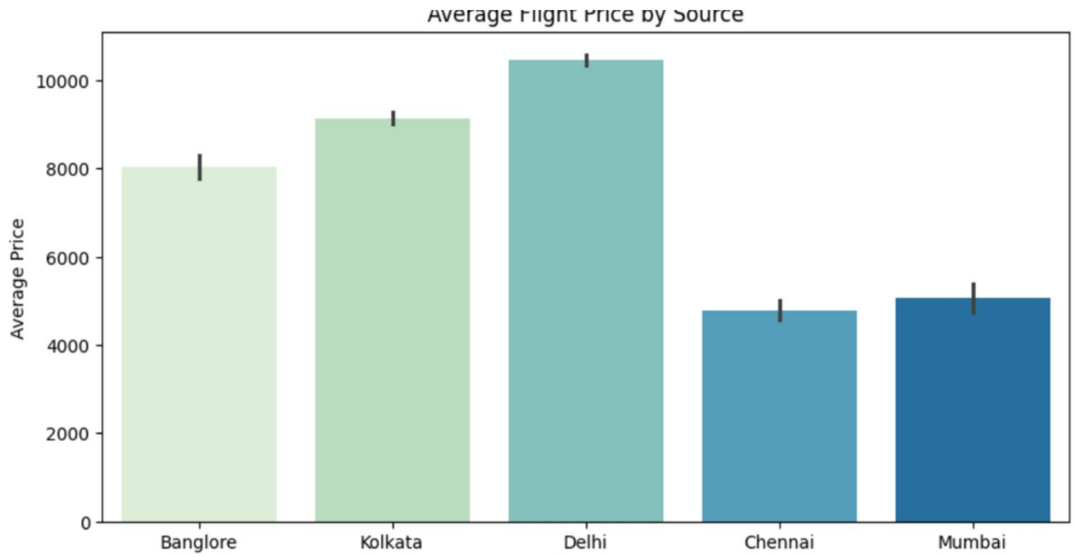

Figure 6 · Average Price by Source

Departure city matters too — flights originating from Delhi average the highest prices, while Chennai departures are the cheapest.

Departure Hour Patterns

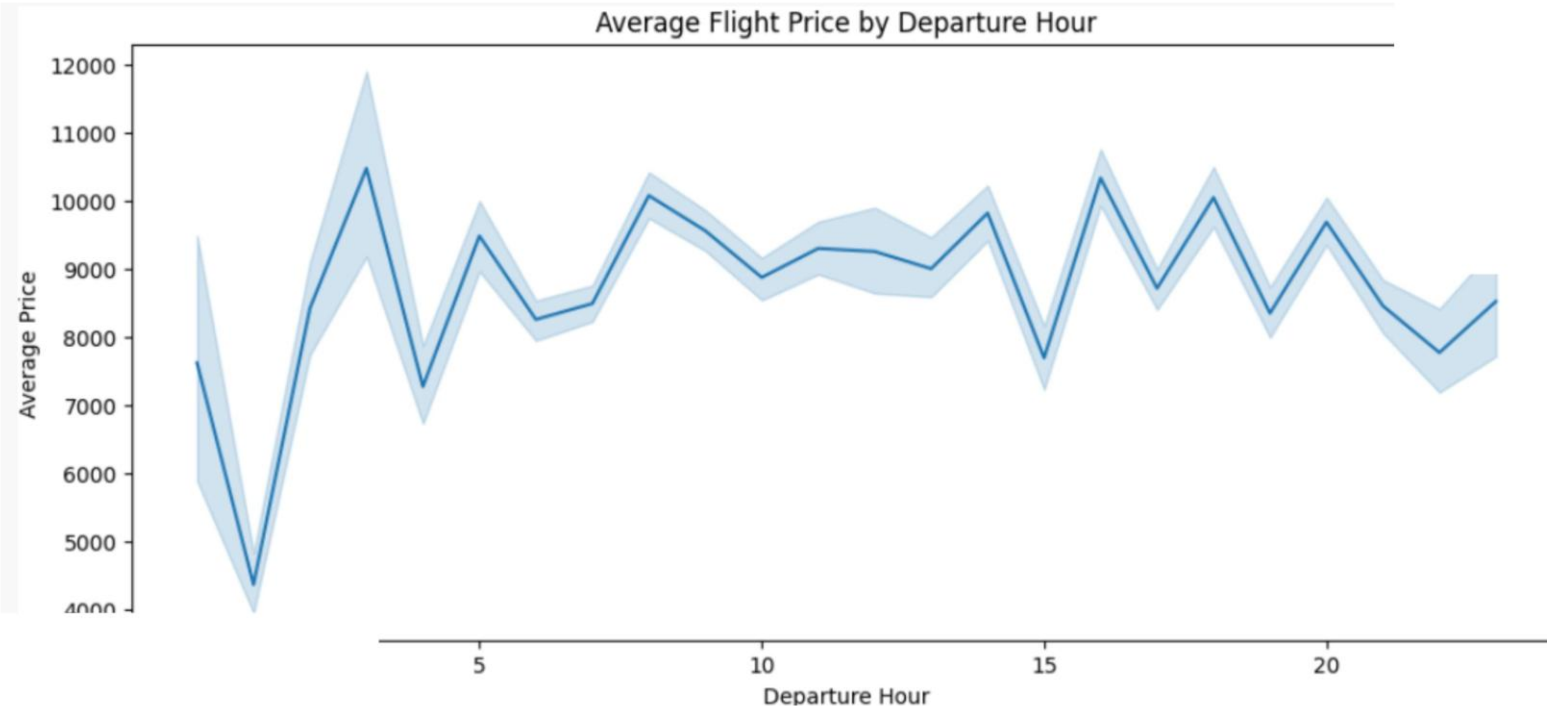

Figure 7 · Average Flight Price by Departure Hour

Prices dip sharply for the 1 AM red-eye slot and peak around early-morning (3 AM) and evening (4–6 PM) departures.

Why It Matters

Departure time shows a visible association with ticket prices, suggesting travelers can find better fares by adjusting flight timing.

Combined with airline, stops, and duration, departure hour rounds out the model's key predictive features.

PREPARING FOR MACHINE LEARNING

Categorical Encoding

Categorical features were converted into numerical representations so the regression model could use them.

Ordinal Encoding

Total_Stops was mapped to ordered numeric values (0–4) to preserve the natural order of stops.

One-Hot Encoding

Airline, Source, and Destination were converted to binary vectors (drop_first=True) — avoiding artificial rank or redundant dummies.

Figure 8 · Ordinal Mapping for Total_Stops

```
df['Total_Stops'] = df['Total_Stops'].replace({
    'non-stop': 0,  '1 stop': 1,  '2 stops': 2,
    '3 stops': 3,  '4 stops': 4
})
```

Figure 9 · One-Hot Encoding

```
categorical_cols = ['Airline', 'Source', 'Destination']
df_encoded = pd.get_dummies(
    df, columns=categorical_cols,
    drop_first=True, dtype=int)
```

Machine Learning Model

Model Used: Linear Regression

Dataset Split

80%	20%
Training Data	Testing Data

Why an 80/20 split?

It provides sufficient data for model optimization while reserving a representative, unseen portion for unbiased performance evaluation.

Figure 10 · Train, Fit & Predict

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
print(y_pred[:10])
```

SUMMARY

Key Findings

01 Flight prices vary meaningfully across airlines and destinations.

02 Number of stops and flight duration are strongly associated with ticket price differences.

03 Average prices also vary by departure source and departure time.

04 Careful data cleaning and feature engineering prepared the dataset for reliable analysis and modeling.

05 A Linear Regression model was trained to predict flight prices from the selected features.

Thank You

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Dataset

kaggle.com/code/muhammadfaizan65/flight-price-prediction/input

Code Notebook

colab.research.google.com/drive/1xFSKQQLEaql7EsP0WKFhVDSV8LPnhSMV